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# Paleomagnetism of the Deschambault Pegmatites: Stillstand and Hairpin at the End of the Paleoproterozoic Trans-Hudson Orogeny, Canada

**D. T. A Symons[1], T. B. Symons[1] and M. T. Lewchuk[2]**

[1]Earth Sciences, University of Windsor, Windsor, ON, N9B 3P4, Canada
[2]School of Geology and Geophysics, University of Oklahoma, Norman, OK, 73019-0628, U.S.A.

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**Abstract.** The 1766 ± 5 Ma Deschambault pegmatites are anorogenic intrusions emplaced into the Glennie domain at the end of the Trans-Hudson Orogeny (THO) in north-central Saskatchewan. They are composed mainly of orthoclase and quartz with minor biotite, muscovite, tourmaline and beryl. A coherent primary characteristic remanence is retained in all 18 sites (170 specimens) that resides in magnetite, hematite and minor pyrrhotite, giving a direction of Dec. = 28.3°, Inc. = 82.1°, $\alpha_{95}$ = 4.0°, and k = 77.5 based on alternating field and thermal step demagnetization and saturation remanence analyses. The pegmatites' pole position, along with recently published ~1810 ± 10 Ma and ~1795 ± 15 Ma poles for the THO, define a stillstand and hairpin in the apparent polar wander path for the THO that marks continent-continent collision of the Archean Superior, Sask and Hearne (?) cratons.
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## 1 Introduction

The Trans-Hudson Orogen (THO) was formed during the Paleoproterozoic when the Archean Hearne and Superior cratons converged in a continent-continent collision, enclosing the THO in between (Hoffman, 1988; Lewry and Collerson, 1990; Lewry et al., 1994) (Figure 1). Accordingly, the evolution of the THO is geologically complex. From a tectono-stratigraphic perspective it is composed of a collage of oceanic plateau/oceanic island, ocean floor, juvenile oceanic arc, continental margin arc, forearc and back-arc basin, continental craton, and collisional intrusive terranes that have been collapsed into an unroofed accretionary prism (*e.g.*, Lewry, 1981; Green et al., 1985; Bickford et al., 1990; Zwanzig et al., 1990; Meyer et al., 1992; Ansdell et al., 1995; Lucas et al., 1996).

*Correspondence to:* D.T.A. Symons, Earth Sciences, University of Windsor, Windsor, ON, N9B 3P4, Canada, dsymons@uwindsor.ca

Through LITHOPROBE seismic and electromagnetic studies, a great deal has been learned about the present subsurface geometry of the various terranes (*e.g.*, Lewry et al., 1994; Lucas et al., 1994; White et al., 1994 and 1996; Hajnal et al., 1996; Jones et al., 1997). Realizing that two-thirds of the Earth's present crust has been produced in the last ~150 Ma as seafloor and orogenic belts, it is evident that the ~150 Ma from ~1.92 to 1.77 Ga represented by pre- and post-collisional orogenesis in the THO (Ansdell et al., 1995; Lucas et al., 1996) must similarly have involved a large percentage of the Earth's surface. LITHOPROBE has, therefore, also sponsored a series of paleomagnetic studies aimed at elucidating the pre-collisional geometry of most THO terranes (*e.g.*, Symons 1991; Symons and MacKay, 1999). This study reports paleomagnetic results from pegmatites at Deschambault Lake and integrates them with two other recently reported results (Symons and Harris, 1999; Symons et al., 1999) to examine post-collisional aspects of THO terrane motions.

The THO includes several lithotectonic elements. On the southeastern side is the Thompson fold belt which is the boundary zone of the Superior craton. On the northwestern side are the pericratonic Peter Lake domain and the Wollaston fold belt, which is the boundary zone of the Hearne craton, and the Reindeer zone lies between the two fold belts. Within the Reindeer zone much of the Paleoproterozoic Manikewan Ocean (Stauffer, 1984), which was about ~5500 km wide at ~1855 Ma (Dunsmore and Symons, 1990; Symons, 1991, 1998), subducted at the Manikewan suture along the Rottenstone domain's southeastern boundary. This placed rocks with equatorial paleolatitudes at ~1855 Ma on the northwest side of the suture against those with polar paleolatitudes on the southeast side. The domains to the southeast of the suture in the THO comprise mostly juvenile Paleoproterozoic rocks – arc and basinal volcanic, intrusive and sedimentary rocks – with windows of the underlying Archean Sask craton that have been subjected to varying amounts of collisional deformation, metamorphism and uplift.

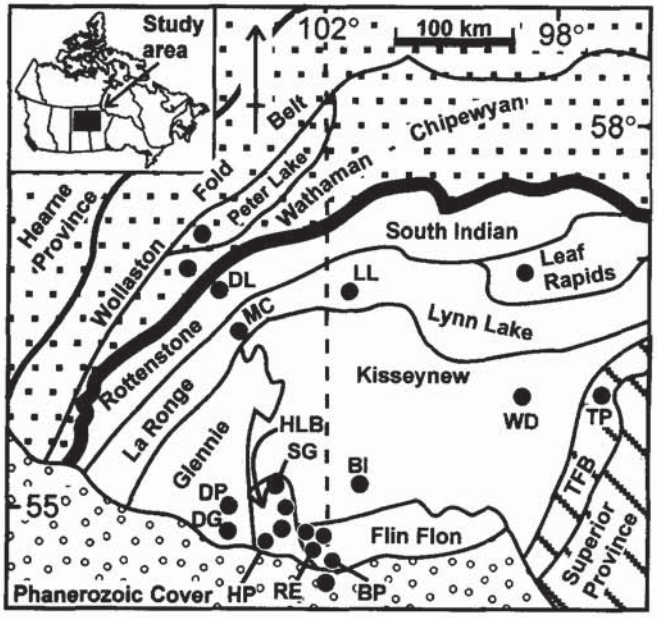



**Fig. 1.** Tectonic elements of the Trans-Hudson Orogen and location of the Deschambault pegmatites. HLB - Hanson Lake block; TFB - Thompson fold belt. Dots show paleomagnetic study locations with those referred to in the text labelled: BI - Big Island gneiss dome; BP - Boot-Phantom intrusive complex; DG - Deschambault granites; DP - Deschambault pegmatites; HP - Hanson Lake pluton; LL - Lynn Lake gabbro pipes; MC - Macoun Lake pluton; RE - Reynard Lake pluton; SG - Sahli granulite; TP - Thompson pegmatites; WD - Wapisu gneiss dome.

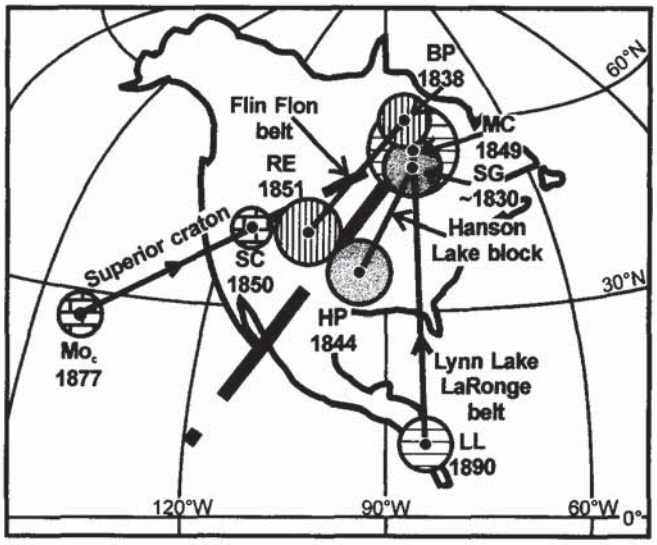



**Fig. 2.** Apparent polar wander path (APWP) segments for the Trans-Hudson Orogen using poles for: Lynn Lake gabbro pipes (LL: Dunsmore and Symons, 1990); Macoun Lake pluton (MC: Symons et al., 1994); Hanson Lake pluton (HP: Gala et al., 1994); Sahli charnockite (SG: Gala et al., 1998); Reynard Lake pluton (RE: Symons, 1995); Boot-Phantom pluton (BP: Symons and MacKay, 1999); Molson dikes' C pole (Mo$_c$: Zhai et al., 1994; Halls and Heaman, 1999); and, Sudbury irruptive complex - norite and granophyre average (Morris, 1984). The wide line denotes the mean 1890 to 1830 Ma trend of the APWP segments that define the "continent-continent" collision in the Trans-Hudson.

Apparent polar wander path (APWP) segments in the ~1890 to ~1830 Ma time span have been determined for the Lynn Lake – La Ronge domain, Hanson Lake block, and Flin Flon domain. These APWP segments along with the segment for the Superior craton project to a common pole at ~1830 Ma, and this is the classical paleomagnetic signature for the accretion of continental blocks in a collisional orogen (Figure 2) (Symons, 1999). What happened in terms of plate motions in the THO from ~1830 Ma to ~1770 Ma then becomes an important question because all of the accreted terranes should share a common APWP. This study of the ~1766 Ma Deschambault pegmatites in the Glennie domain focuses on this question.

First, however, it should be noted that pegmatites have not been a popular target for paleomagnetic studies for obvious reasons. Typically they have a high percentage of quartz and are hard to drill. Once drilled they have a propensity to be brittle and to fracture into pieces upon extraction from their hole because of their large crystal size. In spinner magnetometers, the type most commonly used, they tend to give erratic directions because the magnetic minerals in pegmatites are not uniformly distributed throughout the specimens. Also their coarse crystalline size commonly means that the remanence in pegmatites is often carried by multidomain magnetite or pyrrhotite, resulting in extremely intense and unstable viscous remanent magnetization (VRM) components that totally mask the weaker stable characteristic remanent magnetization (ChRM) components that carry geologic information. Nevertheless, 1760 to 1800 Ma pegmatites are the youngest Paleoproterozoic rocks in the THO. They are mostly post-tectonic and undeformed, and they are found in several domains within the THO (Table 1). Despite the risk of failure, therefore, these pegmatites are an important target.

## 2 Geology

This study was done on the Deschambault pegmatites, an informal name, in the Viney Lake area (NTS 63 L 14) that was mapped geologically by Padgham (1968) at a scale of 1:63 360 (Figure 3). The study area is on the southeast side of the Glennie domain, about 15 km west of the Tabbernor fault that marks the domain's eastern boundary, where the pegmatites are particularly abundant and well exposed in shoreline outcrops around Deschambault Lake. They are the youngest Paleoproterozoic rocks in the study area and in the Glennie domain, and they intrude all Archean and other Paleoproterozoic rock units.

The Deschambault pegmatites occur mainly as regular veins, dikes or sheets up to about two metres thick, but also as pods, screens (sites 1 and 3) and irregular masses (Padgham, 1968). They grade as a continuum from slightly older red to pink granitic pegmatites that are composed mainly of potassium feldspar and quartz with minor biotite to slightly younger pink to white biotite-muscovite-granitic pegmatites with accessory tourmaline and beryl. The oldest red pegmatites have gradational contacts, are late syntectonic structures with a weak to moderate north-northeast-trending shear foliation in places and were not sampled. Pink pegmatites were sampled at sites 2-4, 6, 7, 11, 13 and 15-17. The younger pale pink (sites 1, 8, 9, 12, 14 to 18) to white pegmatites (sites 5 and 10) are post-tectonic with sharp contacts that define north-northeastward

**Table 1.** Summary of radiometric age dates in millions of years.

| Domain | Pegmatites/Veins | Felsic Plutons |
|---|---|---|
| Hearne craton | 1783 ± 47 Uzi, 15 | |
| Wollaston fold belt | 1773 ± 2 Pzi, 03 | 1788 ± 2 Usp, 02 |
| | 1793 ± 4 Umz, 02 | 1779 ± 2 Usp, 02 |
| | | 1765 ± 30 Rwr, 05 |
| | | 1752 ± 41 Kbi, 16 |
| La Ronge belt | 1769 ± 21 Uzi, 07 | 1730 ± 30 Rwr, 05 |
| Glennie domain | 1770 ± 3 Umz, 08 | 1768 ± 2 Umz, 08 |
| | 1770 ± 2 Umz, 08 | |
| | 1770 ± 15 Ual, 08 | |
| | 1762 ± 1 Umz, 08 | |
| Hanson Lake block | 1760 ± 20 Rwr, 09 | 1750 ± 20 Rwr, 05 |
| | | 1773 ± 9 Uzi, 06 |
| Flin Flon belt | 1790 ± 60 Kmu, 17 | 1790 ± 2 Usp, 10 |
| | 1763 ± 55 Rwr, 12 | |
| | 1741 ± 55 Rwr, 12 | |
| | 1735 ± 55 Kwr, 19 | |
| | 1791 ± 4 Amu, 11 | |
| Kisseynew domain | 1783 ± 11 Uzi, 04 | 1799 ± 2 Umz, 01 |
| | 1767 ± 1 Uzi, 04 | 1801 ± 2 Umz, 01 |
| | 1797 ± 1 Umz, 04 | 1799 ± 3 Umz, 01 |
| | 1806 ± 2 Umz, 04 | |
| Thompson fold belt | 1786 ± 3 Uzi, 13 | 1755 ± 60 Kbi, 18 |
| | 1771 ± 2 Uzi, 14 | |

Abbreviations: A = $^{40}Ar/^{39}Ar$; K = K/Ar; P = $^{207}Pb/^{206}Pb$; R = Rb/Sr; U = U/Pb; al = allanite; bi = biotite; mu = muscovite; mz = monazite; sp = sphene (titanite); wr = whole rock; zr = zircon. References: 01 – Ansdell and Norman (1993); 02 – Annesley et al. (1992); 03 – Annesley and Madore (1997); 04 – Ashton et al. (1992); 05 – Bell and Macdonald (1982); 06 – Bickford et al. (1986); 07 – Bickford et al. (1992); 08 – Chiarenzelli (1989); 09 – Coleman (1970); 10 – David et al. (1993); 11 – Fedorowich et al. (1991); 12 – Josse et al. (1974); 13 – Krogh et al. (1985); 14 – Machado et al. (1987); 15 – Orrell and Bickford (1989); 16 – Stevens et al. (1982); 17 – Wanless et al. (1965); 18 – Wanless et al. (1966); 19 – Wanless et al. (1967).

strikes ( N35°±10°E ) and steep to vertical dips. The foliation in the red pegmatites and the preferred orientation of the pink and white pegmatites suggests that they are genetically related to the parallel Tabbernor fault system.

In the study area the Deschambault pegmatites intrude upper amphibolite facies grade schists and gneisses of the Paleoproterozoic Oskikebuk Lake Group that predate D2 east-west thrust structures (McNicoll et al., 1992). They also intrude late syntectonic to post-tectonic granites of the Deschambault Post and "Tower Island" plutons that postdate D3 north-south open fold structures found in the schists and gneisses (Figure 3). Further there is no evident preferred spatial relationship between these plutons and the pegmatites (Padgham, 1968).

The pegmatites have not been dated at Deschambault Lake. Elsewhere in the Glennie domain they have been dated at 1770 ± 3, 1770 ± 2, 1770 ± 15 and 1762 ± 1 Ma by Chiarenzelli (1989) using the U-Pb method on monazite or allanite (Table 1). These pegmatites are similar in composition, northerly structural trend, and in degree of deformation to those found on Deschambault Lake (Forsythe, 1981; Chiarenzilli, 1989). Geologic and LITHOPROBE seismic reflection data suggest strongly that the Hanson Lake block and Glennie domain are part of the same original plate with the former having been more highly metamorphosed and deeply eroded from differential

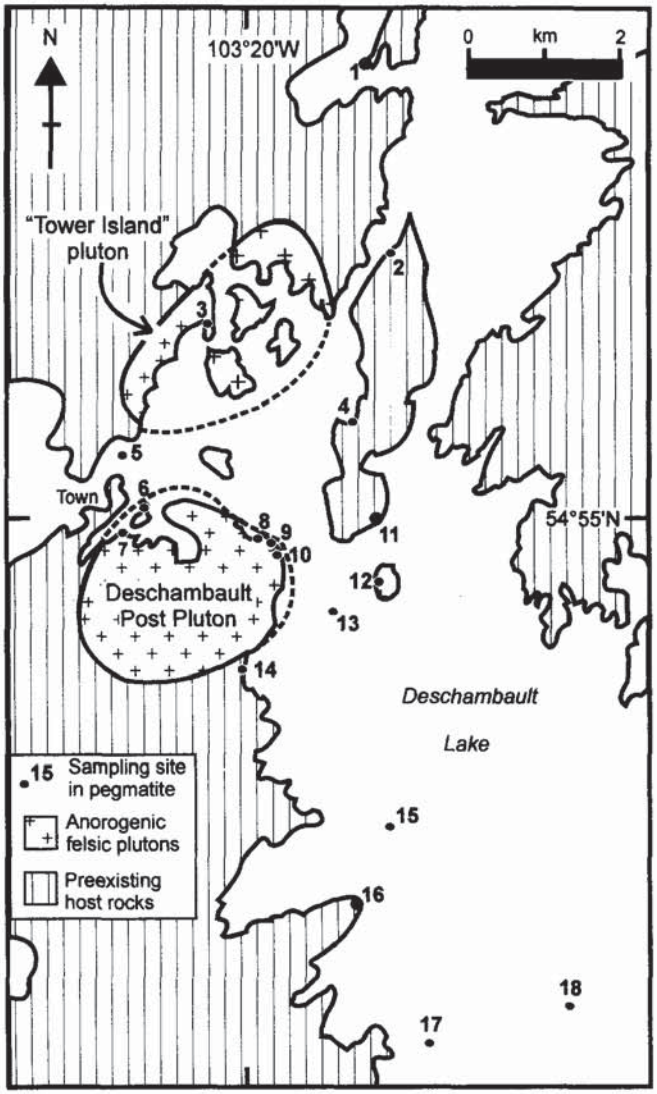



**Fig. 3.** General geology of the Deschambault Lake area simplified from Padgham (1968) with the pegmatite sampling site localities.

uplift on the Tabbernor fault system (Lewry et al., 1994; Maxeiner et al., 1995; Hajnal et al., 1996). Thus the Rb-Sr whole rock age of 1760 ± 20 Ma reported by Coleman (1970) for a pegmatite in the Hanson Lake block is also relevant. The mean and root mean square error for the five pegmatite ages is 1766 ± 5 Ma which is deemed to be their emplacement and magnetization age. This age is consistent with: 1) the 1768 ± 2 Ma age found by Chiarenzelli (1989) for a late granite pluton in the Glennie domain (Table 1); 2) a U-Pb zircon age of 1773 ± 9 Ma on a late granite in the Hanson Lake block (Bickford et al., 1986); 3) U-Pb titanite ages of 1819 ± 7 and 1825 ± 5 Ma for two plutons with D3 fabrics in the Brownell Lake area, some 25 km west of the study area (McNicoll et al., 1992); 4) post-peak metamorphic cooling ages in the Glennie domain of 1780 ± 2, 1773 ± 4 and 1773 ± 2 Ma from U-Pb titanite dating of gneisses (Chiarenzelli, 1989); and, 5) post-peak metamorphic uplift ages in the Hanson Lake block of 1808 ± 3 and 1804 +5/-4 Ma from U/Pb monazite and titanite (Heaman et al., 1994).

## 3 Methods

A total of 18 pegmatite veins were sampled from shoreline outcrops and small islands over an area of ~15 km x 5 km of Deschambault Lake (Figure 3). At each site six 2.54 cm diameter cores were drilled and oriented *in situ* using a sun compass. The veins are quartz-rich so that drilling proved difficult in most cases. Six to 13 specimens per site having a length of 2.20 cm were sliced from these cores (Table 1). They were stored in a magnetically-shielded room with an ambient field of <100 gamma ($\gamma$) for about two months in order to allow their VRM components to decay. All subsequent measurements were done without taking the specimens from the room. The natural remanent magnetization (NRM) of each was measured using an automated Canadian Thin Film CTF-420 cryogenic magnetometer with a noise level for effective resolution of a specimen's magnetization of $< 4 \times 10^{-6}$ A/m. For each site two specimens were demagnetized in an alternating field (AF) in 11 steps up to 120 mT using a Sapphire Instruments SI-4 AF demagnetizer and two more specimens were thermally demagnetized in 11 steps up to 660°C using a Magnetic Minerals MMTD-1 oven. Based on the results from these pilot specimens for each site, the remaining specimens at the site were AF demagnetized in four steps from 20 to 40 - 80 mT. Specimens from 11 sites were further demagnetized at 350°C, and additional AF steps were added for some of the remaining sites. Eight specimens that displayed a range of behaviours were subjected to saturation isothermal remanent magnetization (SIRM) analysis to better identify the magnetic mineral carriers. They were magnetized in a direct field in 10 steps up to 900 mT using a Sapphire Instruments SI-6 pulse magnetizer and then AF demagnetized in six steps to 50 mT. The ChRM's were identified using visual examination of orthogonal vector component plots (As and Zijderveld, 1958) and the principal component method of Kirschvink (1980). Site and unit mean ChRM directions were obtained using Fisher (1953) statistics.

## 4 Results

A total of 170 pegmatite specimens were analysed paleomagnetically (Table 2). Their median NRM intensity was $4.5 \times 10^{-4}$ A/m with lower and upper quartiles of 2.3 and $8.0 \times 10^{-4}$ A/m, respectively. The specimens showed a range of behaviours on AF step demagnetization with a few showing virtually no NRM intensity decay up to 120 mT (Figure 4a), but with most showing between ~30% and ~70% decay (Figures 4b and 4c). Their ChRM component direction was well isolated in most cases over a broad range of coercivities above ~20 mT after the VRM had been removed. The ChRM is typically directed steeply downwards to the north-northeast. A few specimens were found to have antipodal directions (Figure 4d).

Thermal step demagnetization demonstrated that the magnetic mineralogy was heterogeneous. A few specimens showed sharp intensity drops at about 300° ± 25°C, which is the diagnostic unblocking temperature range of pyrrhotite (Figure 5a). Many specimens showed a distinct intensity drop in the 530° to 590°C range that is typical of magnetite (Figure 5b). Still other specimens had distinct intensity drops between ~600° and 675°C to confirm that hematite was present (Figure 5c). Most of the pegmatite specimens disintegrated between about 550° and 650°C. In addition to

**Table 2.** Site mean remanence directions.

| Site No. | Number of Specimens | | AF Demag. | | Th Demag. | | Mean Direction | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | min. | max. | min. | max. | Dec. | Inc. | $\alpha_{95}$ | k |
| | M | E | (mT) | (mT) | (°C) | (°C) | (°) | (°) | (°) | |
| 1 | 10 | 10 | 20 | 80 | 240 | 630 | 23.5 | 70.2 | 10.6 | 21.7 |
| 2 | 9 | 9 | 20 | 120 | 240 | 500 | 66.3 | 86.5 | 8.4 | 38.6 |
| 3 | 11 | 11 | 20 | 70 | 240 | 530 | 94.4 | 86.7 | 7.3 | 39.8 |
| 4 | 7 | 7 | 20 | 120 | 240 | 350 | 42.8 | 82.6 | 10.2 | 36.2 |
| 5 | 6 | 6 | 20 | 80 | 240 | 500 | 143.4 | 85.9 | 12.5 | 29.9 |
| 6 | 10 | 10 | 20 | 120 | 240 | 560 | 42.5 | 71.4 | 10.9 | 20.8 |
| 7 | 11 | 11 | 20 | 60 | 240 | 350 | 65.5 | 84.2 | 8.0 | 34.0 |
| 8 | 13 | 13 | 20 | 60 | 240 | 560 | 1.2 | 72.5 | 6.2 | 45.7 |
| 9 | 10 | 9 | 20 | 50 | 240 | 560 | 299.3 | 80.6 | 21.8 | 6.6 |
| 10 | 13 | 12 | 20 | 120 | 240 | 530 | 30.8 | 76.1 | 8.4 | 28.0 |
| 11 | 11 | 11 | 20 | 120 | 240 | 590 | 70.8 | 81.2 | 9.0 | 26.7 |
| 12 | 11 | 11 | 20 | 120 | 240 | 590 | 16.1 | 74.4 | 7.2 | 41.7 |
| 13 | 7 | 7 | 20 | 70 | 240 | 500 | 8.8 | 76.0 | 8.9 | 46.6 |
| 14 | 10 | 10 | 20 | 120 | 240 | 590 | 60.2 | 84.5 | 11.5 | 18.5 |
| 15 | 6 | 6 | 20 | 80 | 240 | 500 | 70.7 | 72.6 | 12.1 | 31.6 |
| 16 | 8 | 8 | 20 | 120 | 240 | 350 | 304.5 | 76.1 | 15.7 | 13.4 |
| 17 | 8 | 8 | 20 | 120 | 240 | 590 | 114.7 | 85.5 | 10.3 | 29.9 |
| 18 | 9 | 9 | 20 | 120 | 240 | 560 | 358.4 | 82.2 | 8.1 | 41.6 |
| Mean | 18 sites | | | | | | 28.3 | 82.1 | 4.0 | 77.5 |

Notes: Number of specimens measured (M) and giving accepted endpoints (E); range of alternating field (AF) and thermal (Th) demagnetization (Demag.) steps included in the mean from minimum (min.) to maximum (max.) in milliTesla (mT) or degrees Celsius (°C); Mean Direction defined by its declination (Dec.), inclination (Inc.) and radius of cone of 95 % confidence ($\alpha_{95}$) in degrees (°) and precision parameter (k) of Fisher (1953).

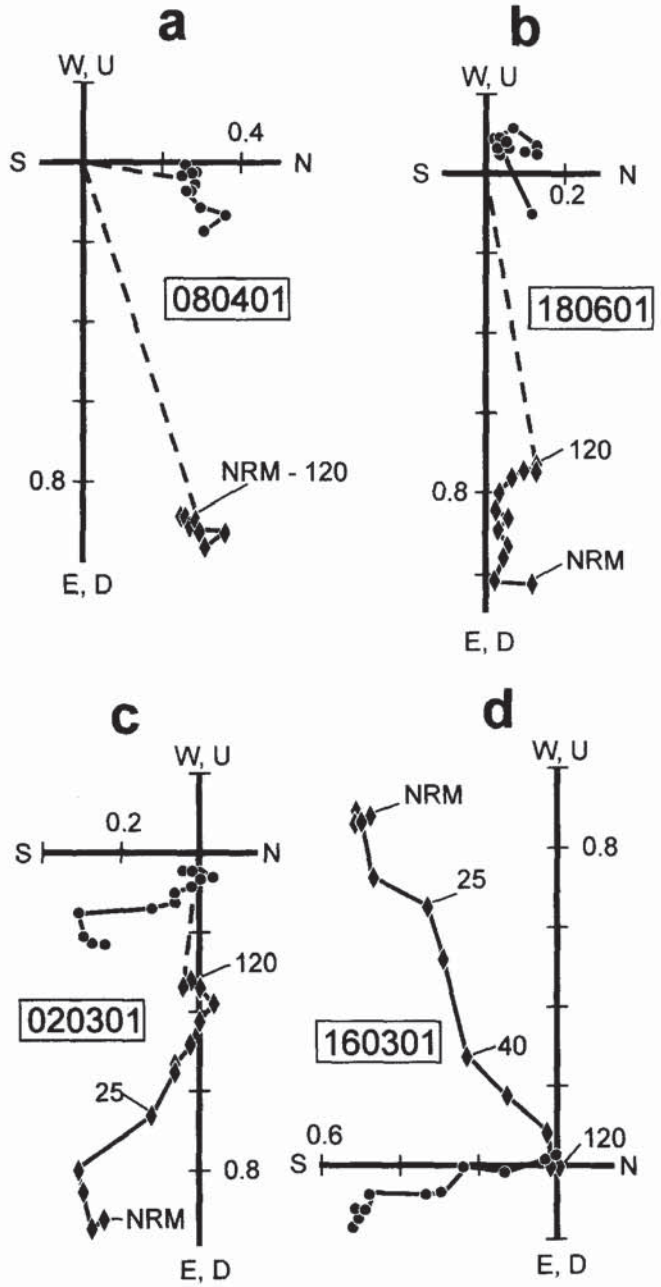



**Fig. 4.** Orthogonal vector plots showing example specimens on alternating field step demagnetization as described in the text. Axes are north (N), east (E), south (S), west (W), up (U), and down (D). Axial values are expressed as a ratio of the specimen's natural remanent magnetization intensity. Projections in the horizontal and vertical planes are circles and diamonds, respectively. Some step fields are labelled in milliTesla.

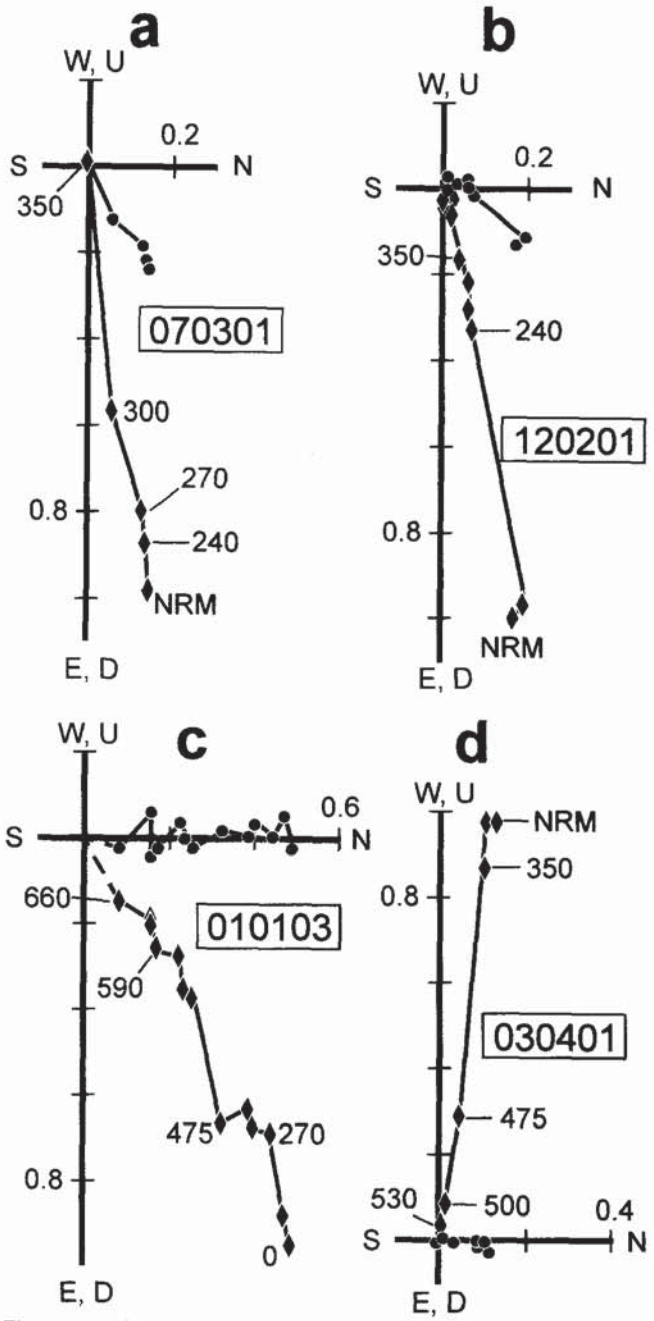



**Fig. 5.** Orthogonal vector plots of example specimens on thermal demagnetization as described in the text. Plotting conventions as in Fig. 4 except some steps are labelled in degrees Celsius.

such definitive specimens, others had two distinct drops suggesting the presence of both pyrrhotite and magnetite, or of both magnetite and hematite. Again the isolated ChRM direction for all these minerals was typically steeply downward to the north-northeast. A number of specimens showed sharp intensity increases with erratic directions either at ~450°C or ~600°C to show that oxidation reactions, most likely of iron sulphides, were forming new magnetic minerals in the specimens. Finally a few specimens, once again, yielded antiparallel directions (Figure 5d). Insufficient specimens were found in which the magnetite, hematite and pyrrhotite directions could be uniquely determined to average to see if a cooling trend was present.

SIRM analysis confirmed the complexity of the pegmatites' magnetic mineralogy. A couple of curves showed rapid acquisition rates to reach saturation by ~300 mT, thus indicating the presence of magnetite and/or pyrrhotite (Figure 6). AF demagnetization of their SIRM indicates single to pseudosingle domain size. Three other specimens proved to have much slower acquisition rates that are indicative of either goethite or hematite. A few of the thermally demagnetized specimens show substantial intensity drops below 240°C where goethite is

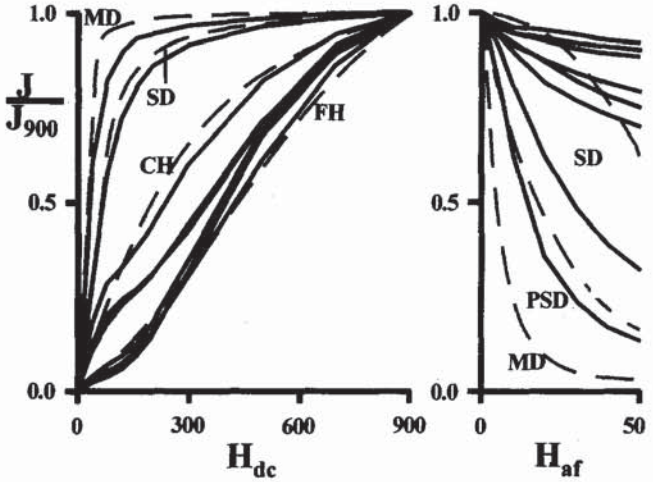



**Fig. 6.** Isothermal remanent magnetization (IRM) curves for example specimens showing (a) acquisition of IRM in a direct field ($H_{dc}$) in milliTesla, and (b) demagnetization of IRM in an alternating field ($H_{af}$) in milliTesla. The intensity (J) is expressed as a ratio of the saturation intensity at 900 mT ($J_{900}$). Shown are typical curves for single, pseudosingle and multidomain magnetite (SD, PSD, MD) and for fine- and coarse-grained hematite (FH, CH).

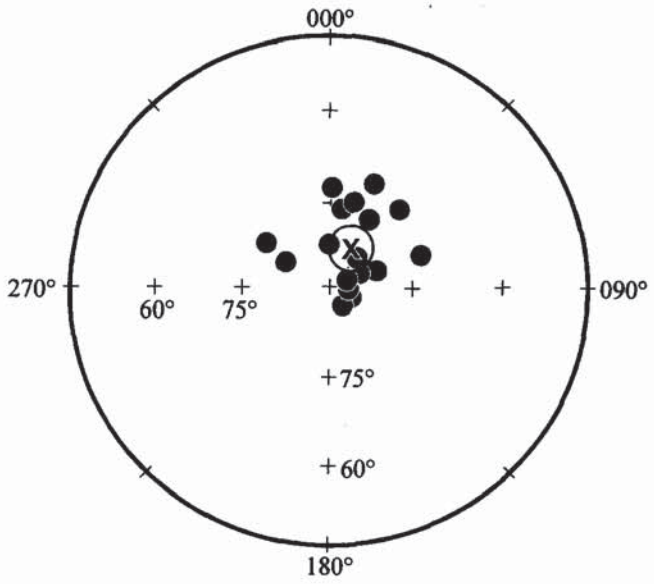



**Fig. 7.** Central portion of an equal-area stereogram showing the site mean remanence directions with the unit mean (X) and its cone of 95% confidence.

preferentially demagnetized so that it is possible that some goethite is present. Nevertheless, the fact that many of the specimens still have a significant undemagnetized remanence intensity above 590°C shows that hematite is a major remanence carrier in many of these pegmatites. The remaining three specimens showed an initial rapid rise in SIRM intensity that is indicative of magnetite or pyrrhotite followed by a slow rise that is indicative of hematite.

Overall the pegmatite specimens were found to be reliable ChRM recorders. Note that only two specimens gave aberrant directions that led to their exclusion from use in calculating the site mean ChRM directions (Table 2). Except for site 9, which is loosely clustered but gives the same mean direction, all of the pegmatite sites gave reasonably well clustered directions with a median precision parameter (k, Fisher, 1953) of 31.6. Similarly the 18 site mean directions are well clustered, giving a unit mean of direction of declination (Dec.) of 28.3°, inclination (Inc.) of 82.1°, radius of the cone of 95% confidence ($\alpha_{95}$) of 4.0°, and k of 77.5 (Figure 7).

## 5 Discussion

This study of the Deschambault pegmatites is the last of five studies undertaken in the THO on rocks in the ~1830 to ~1760 Ma time span, and it is the most definitive. Two of the five studies failed. A paleomagnetic test of 16 pegmatite dikes in the Thompson and Pipe mine pits of INCO near Thompson, Manitoba, in the Thompson fold belt on the margin of the Superior craton failed because these pegmatites failed to retain a stable remanence although apparently useful geologic information was found in several contact tests leading to a second attempt in progress (Figure 1; Harris et al., 1998). Also amphibolitic gneisses from around the Big Island gneiss dome at Kississing Lake on the southern flank of the Kisseynew domain had very large viscous remanent magnetization components and, although the individual specimens had well-defined ChRM components, their directions were poorly clustered for most sites. Further, the site mean ChRM directions, while clearly post-folding and metamorphic in origin, did not form a cluster from which to derive a useful pole position (Symons and Harris, 1998).

A paleomagnetic study on the Wapisu gneiss dome in the northern part of the Kisseynew domain proved successful (Figure 1) (Symons and Harris, 1999). This dome is nearly circular and ~6 km in diameter. It is composed of amphibolitic gneisses derived from turbiditic sediments and sill-like leucogranitic intrusions (Baldwin et al., 1979). Based on data from 17 sites around the perimeter in the steeply-dipping flanks, a stable post-folding ChRM direction was isolated that gives a pole position at 119.5° W, 75.8° N ($d_p$ = 9.5°, $d_m$ = 10.0°) (Figure 8). Based on radiometric ages for metamorphic and post-metamorphic rocks in the area, the Wapisu gneiss dome's ChRM was interpreted to record post-peak metamorphic cooling at ~1810 ± 10 Ma. Assuming that exhumation occurred without tilt, it was speculated that the pole position marked an ~90° hairpin in the APWP from the juvenile terrains of the THO, marking the collisional impact of the Archean Sask craton and/or Hearne craton against the Paleoproterozoic Kisseynew domain and driving it into the Archean Superior craton.

A positive paleomagnetic result was obtained from the Deschambault Post and "Tower Island" granite plutons in the Glennie domain that are intruded by the Deschambault pegmatites of this study (Figures 1 and 3) (Symons et al., 1999). These small circular to ovate late syntectonic to post-tectonic intrusions are composed of massive leucocratic granodiorite-granite and quartz monzonite and outcrop over ~4.3 and ~5.8 $km^2$, respectively (Padgham,

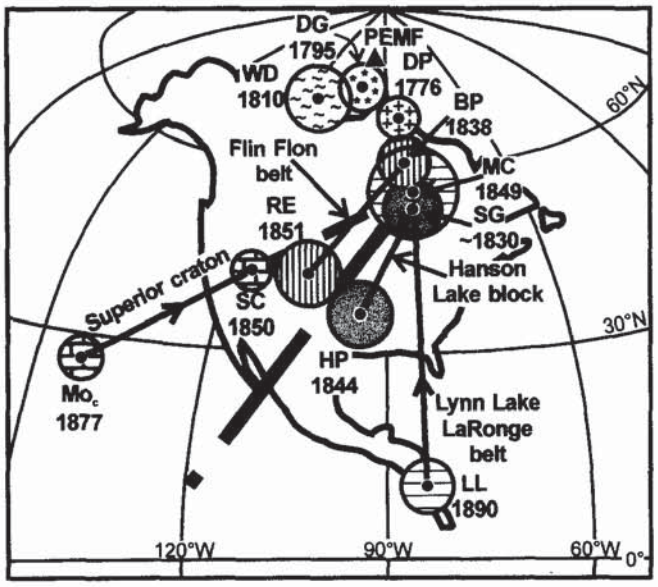



**Fig. 8.** Pole positions for the Trans-Hudson Orogen showing the 1890 - 1830 Ma apparent polar wander path segments from Fig. 2 as well as the post-assembly 1830 - 1766 Ma poles for the ~1810 ± 10 Ma Wapisu gneiss dome (WD: Symons and Harris, 1998) in the Kisseynew domain (Figure 1), the ~1795 ± 15 Ma Deschambault granites (DG: Symons et al., 1999), and the 1766 ± 5 Ma Deschambault pegmatites (DP) pole from this study. The triangle is the present Earth's magnetic field pole (PEMF).

1968). Neither pluton has been reliably dated radiometrically. Geologists mapping in the Glennie domain note that the only period of granodiorite-granite magmatism in the domain occurred between ~1830 and 1840 Ma, and that these plutons appear similar to ones found ~25 km to the west in the Brownell Lake area that contain D2 and D3 fabrics that imply an age of ~1830 Ma (McNicoll et al., 1992; G. Delaney, pers. comm., 1999). Symons et al. (1999) suggested an emplacement age of ~1795 ± 15 Ma based on: (a) the lack of reported fabric by Padgham (1968), implying an age of <1822 ± 6 Ma; (b) the undistorted circular shape of the Deschambault Post pluton that is incompatible with the steeply-plunging D3 syncline in the Oskikebuk Lake schists and gneisses into which it is intruded; (c) peak metamorphic U-Pb monazite ages of 1808 ± 3 Ma and 1804 +5/-4 Ma in the Hanson Lake area about 25 km to the southeast; (d) cross-cutting 1766 ± 5 Ma pegmatites of this study that provide a minimum age; and, (e) the fact that the pole positions for the plutons, if corrected for the plunge of the D3 syncline found in the host rocks, would be radically different from all other THO poles. The combined pole position determined from 14 sites in the Deschambault Post pluton and 6 sites in the "Tower Island" pluton is located at 102.4° W, 76.7° N ($d_p$ = 5.9°, $d_m$ = 6.2°). Symons et al. (1999) noted that the pole fits reasonably well on the extrapolated extensions of the apparent polar wander path segments for the Lynn Lake – La Ronge domain and for the Hanson Lake block, which bound the Glennie domain. Thus they suggested that the pole may be indicating a substantial slowdown in the rate of apparent polar wander or drift rate that is a common feature found for continent-continent collisions.

The pole position derived from the Deschambault pegmatites (DP) is located at 84.0° W, 67.5° N ($d_p$ = 7.6°, $d_m$ = 7.8°) (Figure 8). On calculation this pole is significantly different at the 95% confidence level from the pole for the intruded ~1795 ± 15 Ma Deschambault Post and "Tower Island" granite plutons (DG) using the McFadden and Lowes (1981) discrimination test. This difference implies a very few degrees of either polar wander or terrane tilt between ~1795 and ~1766 Ma, and it provides a paleomagnetic contact test (Everitt and Clegg, 1962) that implies that the ChRMs of the pegmatites are primary. The presence of specimens with a reversed polarity and the high unblocking temperatures in both magnetite and hematite similarly support the retention of a primary remanence despite the pole's closeness to the present Earth's dipole field (Figure 8). Note also that these unblocking temperatures are close to the closure temperature for the U-Pb monazite radiometric system of about 695° ± 45°C (Kingsbury et al., 1993) so that the 1766 ± 5 Ma age is closely matched by the ChRM.

Further, both the DP and DG poles lie on the extensions of the APWP segments for the Lynn Lake – La Ronge belt and the Hanson Lake block, and they are the two terranes that bound the Glennie domain on its west and east sides, respectively (Figure 1). This geometry implies a swing of ~40° counterclockwise in the direction of apparent polar wander and thus in terrane drift direction. This result is entirely consistent with the ~90° counterclockwise swing or hairpin recorded by the pole for the ~1810 ± 10 Ma Wapisu gneiss dome pole from the Kisseynew domain in relation to the APWP segments for the bounding Superior craton and Flin Flon belt. The fact that substantial counterclockwise changes in the APWP direction are evidenced for the Glennie and Kisseynew domains, indicates a major change in direction of plate motion between ~1830 to 1775 Ma relative to the previous 60 Ma period. As Irving and Park (1973) noted, such a hairpin is an expected result from an orogenic collision. The North American APWP, for example, records hairpins of ~90° during the ~370 Ma Acadian, ~325 Ma Alleghanian and ~100 Ma Laramide orogenies (Besse and Courtillot, 1991; Van der Voo, 1993).

The third aspect shown by the DP pole is that it marks a prolonged 60 Ma period of very slow apparent polar wander. The average rate of wander for the four defined APWP segments prior to 1830 Ma is about 14 ± 4 cm/a. This is a moderately rapid rate but certainly within the range that has been well defined for large plates over several tens of millions of years during the Phanerozoic that have given values up to ~20 cm/a (Meert et al., 1993). In contrast, the three poles for the THO that postdate ~1830 Ma do not show a coherent age progression and indicate a much slower rate of no more than 5 ± 4 cm/a with the Deschambault pegmatites' pole giving a very slow rate of ~1 cm/a. Thus the post ~1830 Ma paleomagnetic results for the THO define a stillstand or period of slow apparent polar wander, which is a common feature for colliding continents because their drift tends to stall or slow dramatically upon collision with respect to the time-averaged magnetic or rotational pole (Irving and Park, 1973).

The Archean cratons involved in the collision are still uncertain. Given the 1815 ± 15 Ma ages for regional metamorphism in the Kisseynew and Glennie domains, it is clear that both the Sask and Superior cratons are involved. It could be that they collided with each other with the Hearne craton not yet involved or, alternatively, that the Sask and Superior cratons were adjacent to each other and that the Hearne craton impacted against them at ~1815 ± 15 Ma. Needed to help resolve the problem is reliable paleomagnetic data for the 1830 to 1770 Ma period for the Superior and Hearne cratons and/or their bounding Thompson and Wollaston fold belts, respectively, with the ~1770 Ma pegmatites (Table 1) being an obvious target.

## 6 Conclusions

The 1766 ± 5 Ma pegmatites at Deschambault Lake retain a primary ChRM that resides principally in magnetite and hematite but also in pyrrhotite in a few specimens. The ChRM is predominantly of normal polarity, is considered to be a thermoremanent magnetization, forms a coherent population of site mean directions, and gives a pole position of 84.0° W, 67.5° N ($d_p$ = 7.6°, $d_m$ = 7.8°). This result, in concert with previously published poles from the ~1810 ± 10 Ma Wapisu gneiss dome and ~1795 ± 15 Ma Deschambault Post and "Tower Island" plutons, defines a stillstand and hairpin in the APWP for the THO terranes following assembly and accretion at ~1830 Ma through to ~1760 Ma. This pattern of paleomagnetic poles is typically found in a plate during the several tens of millions of years following a collisional orogeny.

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## 6 References

Annesley, I.R., and Madore, C., U-Pb geochronology of peraluminous pegmatites from the Wollaston Lake area, northern Saskatchewan, *Geol. Assoc. Can., An. Mtg., Abst. Vol. 22*, A4, 1997.

Annesley, I.R., Madore, C., and Krogh, T.E., U-Pb zircon, titanite and monazite ages from the Wollaston Domain: A summary, *Saskatchewan Geol. Surv., Misc. Rep. 92-4*, 61-65, 1992.

Ansdell, K.M., and Norman, A.R., The timing of magmatism, deformation and metamorphism along the southern flank of the Kisseynew Domain, *LITHOPROBE Rep. 34*, 219-225, 1993.

Ansdell, K.M., Lucas, S.B., Connors, K.A., and Stern, R.A., Kisseynew metasedimentary gneiss belt, Trans-Hudson Orogen (Canada): back-arc origin and collisional inversion, *Geology, 23*, 1039-1043, 1995.

As, J.A., and Zijderveld, J.D.A., Magnetic cleaning of rocks in palaeomagnetic research, *R. Astron. Soc., Geophys. J., 1*, 308-319, 1958.

Ashton, K.E., Hunt, P.A., and Froese, E., Age constraints on the evolution of the Flin Flon volcanic belt and Kisseynew belt, Saskatchewan and Manitoba, *Geol. Surv. Can. Pap. 91-2*, 55-69, 1992.

Baldwin, D.A., Frohlinger, T.G., and Kendrick, G., Geology of the Nelson House-Pukatawagan Region (Burntwood Project), *Manitoba Min. Res. Div., Geol. Maps 78-3-12 to 78-3-22 (in Geol. Rep. 78-3)*, 1979.

Bell, K., and Macdonald, R., Geochronological calibration of the Precambrian Shield in Saskatchewan, *Saskatchewan Geol. Surv., Misc. Rep. 82-4*, 17-22, 1982.

Besse, J., and Courtillot, V.E., Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 100 Ma, *J. Geophys. Res., 96*, 4029-4050, 1991.

Bickford, M.E., Collerson, K.D., Lewry, J.F., Van Schmus, W.R., and Chiarenzelli, J.R., Proterozoic collisional tectonism in the Trans-Hudson Orogen, Saskatchewan, *Geology, 18*, 14-18, 1990.

Bickford, M.E., Collerson, K.D., Lewry, J.F., and Orrell, S.E., Pegmatites and leucogranites as probes of crust beneath allochthonous orogenic rocks in the Glennie and La Ronge Domains, *Saskatchewan Geol. Surv., Misc. Rep. 92-4*, 124-129, 1992.

Bickford, M.E., Van Schmus, W.R., Macdonald, R., Lewry, J.F., and Pearson, J.G., U-Pb zircon geochronology project for the Trans-Hudson: Current sampling and recent results, *Saskatchewan Geol. Surv., Misc. Rep. 86-4*, 101-107, 1986.

Chiarenzelli, J.R., The Nistowiak and Guncoat gneisses: Implications for the tectonics of the Glennie and LaRonge domains, northern Saskatchewan, Canada, *Ph.D. thesis, Univ. Kansas*, 229p., 1989.

Coleman, L.C., Rb-Sr isochrons for some Precambrian rocks in the Hanson Lake area, Saskatchewan, *Can. J. Earth Sci., 7*, 338-345, 1970.

David, J., Machado, N., Bailes, A.H., and Syme, E.C., U-Pb geochronology of the Flin Flon – Snow Lake belt: new results, *LITHOPROBE Rep. 34*, 84-87, 1993.

Dunsmore, D.J., and Symons, D.T.A., Paleomagnetism of the Lynn Lake gabbros in the Trans-Hudson Orogen and closure of the Superior and Slave cratons, *Geol. Assoc. Can., Spec. Rep. 37*, 215-228, 1990.

Everitt, C.W.F., and Clegg, J.A., A field test of paleomagnetic stability, *R. Astron. Soc., Geophy. J., 6*, 312-319, 1962.

Fedorowich, J., Stauffer, M., and Kerrich, R., Structural setting and fluid characteristics of the Proterozoic Tartan Lake gold deposit, Trans-Hudson Orogen, northern Manitoba, *Econ. Geol., 86*, 1434-1467, 1991.

Fisher, R.A., Dispersion on a sphere, *R. Soc. Lond. Proc., Ser. A, 217*, 295-305, 1953.

Forsythe, L.H., Geology of the Lac La Ronge – Hunter Bay – Cartier Lake area, Saskatchewan, *Saskatchewan Dept. Min. Res. Open File, Rep. 81-1*, 20p., 1981.

Gala, M.G., Symons, D.T.A., and Palmer, H.C., Paleomagnetism and geotectonic implications of the Hanson Lake pluton, Hanson Lake block, Trans-Hudson orogen, *Saskatchewan Geol. Soc., Misc. Rep. 94-4*, 116-122, 1994.

Gala, M.G., Symons, D.T.A., and Palmer, H.C., Geotectonics of the Hanson Lake block, Trans-Hudson orogen, central Canada: a preliminary paleomagnetic report, *Precambrian Res., 90*, 85-101, 1998.

Green, A.G., Hajnal, Z., and Weber, W., An evolutionary model of the western Churchill Province and western margin of the Superior Province in Canada and the north-central United States, *Tectonophysics, 116*, 281-322, 1985.

Hajnal, Z., Lucas, S., White, D., Lewry, J., Bezden, S., Stauffer, M.R., and Thomas, M.D., Seismic reflection images of high-angle faults and linked detachments in the Trans-Hudson Orogen, *Tectonics, 15*, 427-439, 1996.

Halls, H.C., and Heaman, L.M., The paleomagnetic significance of new U-Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba, *Can. J. Earth Sci.*, in press, 1999.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., Peck, D.C., and Turek, A., Tectonic analysis of the accretion of the Trans-Hudson Orogen to the Superior Craton from paleomagnetic and geobarometric corrections, *Manitoba En. Mines Rep. Activities 1998*, 66-68, 1998.

Heaman, L.M., Maxeiner, R.O., and Slimmon, W.L., 1993/94 U-Pb geochronological investigations in the Trans-Hudson Orogen, *Saskatchewan En. Mines, Misc. Rep. 94-4*, 96-100, 1994.

Hoffman, P.F., United plates of America, the birth of a craton, Early Proterozoic assembly and growth of Laurentia, *An. Rev. Earth Planet. Sci., 16*, 543-603, 1988.

Irving, E., and Park, J.K., Hairpins and superintervals, *Can. J. Earth Sci., 10*, 1318-1324, 1973.

Jones, A.G., Garcia, X., Grant, N.J., Ledo, L., and Ferguson, I.J., Regional electrical structure of the Trans-Hudson Orogen, *LITHOPROBE Rep. 62*, 130-146, 1997.

Josse, G., Clark, G.S., and Bailes, A.H., Rubidium-strontium age determinations from the File-Morton-Woosey Lakes area of the Flin Flon volcanic belt, *Centre for Precambrian Studies, An. Rep. 1974*, 40-50, 1974.

Kingsbury, J.A., Miller, C.F., Wooden, J.L., and Harrison, T.M., Monazite paragenesis and U-Pb systematics in rocks of the eastern Mojave Desert, California, U.S.A.: implications for thermochronometry, *Chem. Geol., 110*, 147-167, 1993.

Kirschvink, J.L., The least squares line and plane and the analysis of palaeomagnetic data, *R. Astron. Soc., Geophys. J., 62*, 699-718, 1980.

Krogh, T.E., Heaman, L., Machado, N., and Weber, W., U-Pb geochronology program: Thompson belt and Pikwitonei domain, *Manitoba En. Mines, Rep. Activities 1985*, 183-184, 1985.

Lewry, J.F., Lower Proterozoic arc-midcontinent collisional tectonics in the western Churchill Province, *Nature, 294*, 69-72, 1981.

Lewry, J.F., and Collerson, K.D., The Trans-Hudson Orogen: Extent, subdivisions and problems, *Geol. Assoc. Can. Spec. Rep. 37*, 1-14, 1990.

Lewry, J.F., Hajnal, Z., Green, A., Lucas, S.B., White, D., Stauffer, M.R., Ashton, K.E., Weber, W., and Clowes, R.M., Structure of a Paleoproterozoic continent-continent collision zone: A LITHOPROBE seismic reflection profile across the Trans-Hudson Orogen, Canada, *Tectonophysics, 232*, 143-160, 1994.

Lewry, J.F., Hajnal, Z., Green, A.G., Lucas, S.B., White, D.J., Stauffer, M.R., Ashton, K.E., Weber, W., and Clowes, R.M., Structure of a Paleoproterozoic continent-continent collision zone: A LITHOPROBE seismic reflection profile across the Trans-Hudson Orogen, Canada, *Tectonophysics, 232*, 143-160, 1994.

Lucas, S., White, D., Hajnal, Z., Lewry, J., Green, A., Clowes, R., Zwanzig, H., Ashton, K., Schledewitz, D., Stauffer, M., Norman, A., William, P., and Spence, G., Three-dimensional collisional structure of the Trans-Hudson Orogen, Canada. *Tectonophysics, 232*, 161-178, 1994.

Lucas, S.B., Stern, R.A., Syme, E.C., Reilly, B.A., and Thomas, D.J., Intraoceanic tectonics and the development of continental crust: 1.92-1.84 Ga evolution of the Flin Flon Belt (Canada), *Geol. Soc. Am. Bull., 108*, 602-629, 1996.

Machado, N., Heaman, L., Krogh, T.E., and Weber, W., U-Pb geochronology program: Thompson Belt – northern Superior Province, *Manitoba En. Mines, Rep. Activities 1987*, 145-147, 1987.

Maxeiner, R.O., Sibbald, T.I.I., Lewry, J.F., and Watters, B.R., Geology of the Deschambault-Tulabi-Hanson Lakes area, southern Hanson Lake Block, northern Saskatchewan (part of NTS 63L/10, 11, 14, 15), *Geol. Surv. Can., Open File 3218*, 137p, 1995.

McFadden, P.L., and Lowes, F.J., The discrimination of mean directions drawn from Fisher distributions, *R. Astron. Soc. Geophys. J., 67*, 19-33, 1981.

McNicoll, V.J., Delaney, G.D., Parrish, R.R., and Heaman, L.M., U-Pb age determinations from the Glennie Lake Domain, Trans-Hudson Orogen, Saskatchewan, *Geol. Surv. Can. Paper 92-2*, 57-72, 1992.

Meert, J., Van der Voo, R., Powell, C.M., Li, Z-X, McElhinny, M.W., Chen, Z., and Symons, D.T.A., A plate-tectonic speed limit?, *Nature, 363*, 216-217, 1993.

Meyer, M.T., Bickford, M.E., and Lewry, J.F., The Wathaman batholith: a short-lived Early Proterozoic continental arc in the Trans-Hudson Orogen, Canada, *Geol. Soc. Am. Bull., 104*, 1073-1085, 1992.

Morris, W.A., Paleomagnetic constraints on the magmatic, tectonic, and metamorphic evolution of the Sudbury basin region, *Ontario Geol. Surv., Spec. Vol. 1*, 411-427, 1984.

Orell, S.E., and Bickford, M.E., Geology of Early Proterozoic gneissic rocks, Black Birch Lake area: Progress Report 2, *Saskatchewan Geol. Surv., Misc. Rep. 89-4*, 105-109, 1989.

Padgham, W.A., The geology of the Deschambault Lake district. *Saskatchewan Dept. Mineral Res. Rep. 114*, 92p, 1968.

Stauffer, M.R., Manikewan: An early Proterozoic ocean in central Canada, its igneous history and orogenic closure, *Precambrian Res., 25*, 252-281, 1984.

Stevens, R.D., Delabio, R.N., and Lachance, G.R., Age determinations and geological studies: K-Ar isotopic ages, *Geol. Surv. Can., Pap. 81-2*, 102p., 1982.

Symons, D.T.A., Paleomagnetism of the Proterozoic Wathaman Batholith and the suturing of the Trans-Hudson Orogen in Saskatchewan, *Can. J. Earth Sci., 28*, 1931-1938, 1991.

Symons, D.T.A., Paleomagnetism of the 1851 Ma Reynard Lake pluton, Flin Flon domain, Trans-Hudson orogen: geotectonic implications, *Saskatchewan Geol. Surv., Misc. Rep. 95-4*, 137-144, 1995.

Symons, D.T.A., Precambrian plate tectonic models: Shifting the paleomagnetic paradigm for orogens such as the Trans-Hudson in Canada, *Phys. Chem. Earth, 23*, 753-759, 1998.

Symons, D.T.A., and Harris, M.J., Paleomagnetic data from the Big Island gneiss dome at Kisseynew Lake (NTS 63N/3), *Manitoba En. Mines, Rep. Activities 1998*, 25-31, 1998.

Symons, D.T.A., and Harris, M.J., The ~1830 Ma Trans-Hudson hairpin from paleomagnetism of the Wapisu gneiss dome, Kisseynew domain, Manitoba, *Can. J. Earth Sci.*, in press, 1999.

Symons, D.T.A., Lohnes, C.A., and Lewchuk, M.T., Geotectonics of the Lynn Lake – La Ronge arc in the Trans-Hudson orogen from paleomagnetism of the Paleoproterozoic Macoun Lake pluton, *Saskatchewan Geol. Surv., Misc. Rep. 94-4*, 123-131, 1994.

Symons, D.T.A., and MacKay, C.D., Paleomagnetism of the Boot-Phantom Pluton and the amalgamation of the juvenile domains in the Paleoproterozoic Trans-Hudson Orogen, Canada, *Internat. Conf. Basement Tect., 13*, 313-331, 1999.

Symons, D.T.A., Symons, T.B., and Lewchuk, M.T., Paleomagnetism of post-tectonic Paleoproterozoic plutons in the Viney Lake area (NTS 65L-14), southeastern Glennie domain, Trans-Hudson orogen, *Saskatchewan Geol. Surv., Misc. Rep. 99-4*, 140-147, 1999.

Van der Voo, R., *Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans*, Cambridge Univ. Press, U.K., 1993.

Wanless, R.K., Stevens, R.D., Lachance, G.R., and Rimsaite, J.Y.H., Age determinations and geological studies: Part 1 – Isotopic ages, Report 5, *Geol. Surv. Can. Pap. 64-17*, 125p., 1965.

Wanless, R.K., Stevens, R.D., Lachance, G.R., and Rimsaite, J.Y.H., Age determinations and geological studies: K-Ar isotopic ages, Report 6, *Geol. Surv. Can. Pap. 65-17*, 101p., 1966.

Wanless, R.K., Stevens, R.D., Lachance, G.R., and Edmonds, C.M., Age determinations and geological studies: K-Ar isotopic ages, Report 7, *Geol. Surv. Can. Pap. 66-17*, 120p., 1967.

White, D.J., Lucas, S.B., Hajnal, Z., Green, A.G., Lewry, J.F., Weber, W., Bailes, A.H., Syme, E.C., and Ashton, K.E., Paleoproterozoic thick-skinned tectonics: LITHOPROBE seismic reflection results from the eastern Trans-Hudson Orogen, *Can. J. Earth Sci., 31*, 458-469, 1994.

White, D.J., Lucas, S.B., Hajnal, Z., Zwanzig, H., and Lewry, J.F., Seismic reflection images across the Kisseynew Domain northern boundary, *LITHOPROBE Rep. 55*, 95-98, 1996.

Zhai, Y., and Halls, H.C., Multiple episodes of dike emplacement along the northwestern margin of the Superior Province, Manitoba, *J. Geophys. Res., 99*, 21, 717-21, 732, 1994.

Zwanzig, H.V., Kisseynew gneiss belt in Manitoba: stratigraphy, structure, and tectonic evolution, *Geol. Assoc. Can. Spec. Paper 37*, 95-120, 1990.